Function Approximation Methods for the Inverse Transform Sampling

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Abstract

The inverse transform sampling is a classical sampling technique for generating random numbers using the inverse of the Cumulative Distribution Function. However, analytical solution of the inverse CDF cannot be obtained in most cases, having to resort to a numerical root-finding method, which can suffer from much calculation time. To address this, this study shows a way of constructing a surrogate approximation model for the inverse CDF, using barycentric rational interpolation, under the condition that the the density function is given.

Keywords: Random number generation, Inverse transform sampling, Rational interpolation.

1 Introduction

The inverse transform sampling is a classical, deterministic technique that generates a target joint sample $(X, Y) = (x, y)$ from independent uniform random numbers $(U, V) = (u, v)$ via the inverse of Cumulative Distribution Functions, as $x = F_X^{-1}(u)$, $y = F_{Y|X=x}^{-1}(v)$. In sophisticated financial engineering, alternative approaches such as rejection sampling and Markov Chain Monte Carlo (MCMC) methods face a fundamental limitation: because they rely on random acceptance steps or generate auto-correlated sequences, they cannot be directly integrated with Quasi-Monte Carlo (QMC) simulations. QMC requires deterministic mappings of low-discrepancy sequences to accelerate convergence. Therefore, implementing a fast, deterministic inverse transform sampling framework remains indispensable.

For most practical joint distributions, however, the required inverse CDFs cannot be obtained analytically, forcing one to resort to numerical root-finding algorithms (e.g., Brent's method) for every single sample. This computational bottleneck is particularly acute in multivariate financial applications, such

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as multi-asset option pricing and copula-based tail-risk management, which frequently require real-time sampling from non-linear and asymmetric bivariate distributions. Sequential root-finding paired with numerical integration for millions of samples is computationally prohibitive, posing a critical execution-time bottleneck.

To overcome this challenge, we propose a way of constructing surrogate model for inverse CDFs using Floater-Hormann (FH) barycentric rational interpolation [1, 2]. Unlike polynomial interpolation, which suffer from Runge's phenomenon due to the steep gradients of inverse CDFs, the FH method guarantees pole-free and numerically stable approximants directly from precomputed data. By selecting particularly complex density functions as examples, we aim to gain deeper insights and practical heuristics for approximating intricate inverse CDFs, thereby contributing to the future development of more robust approximation methodologies.

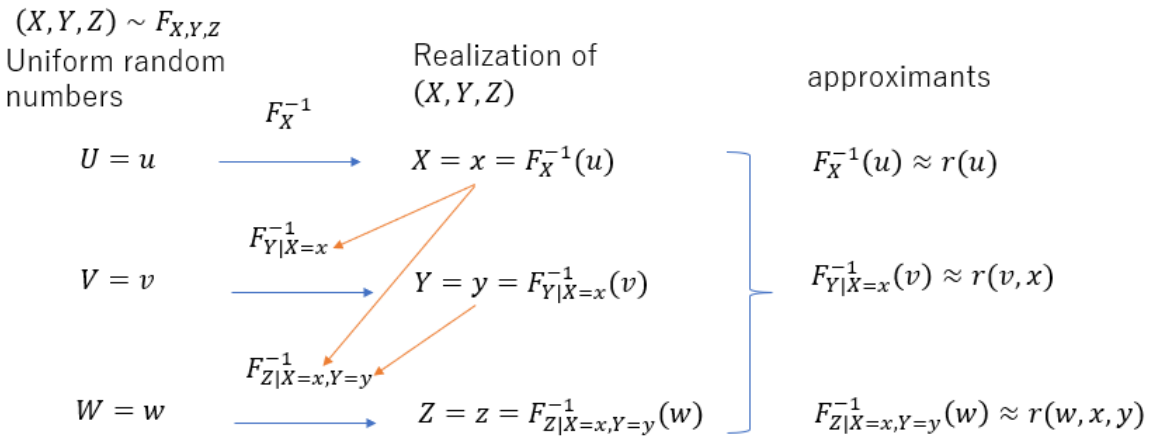


Figure 1: the inverse transform sampling and our strategy

The remainder of this paper is organized as follows: Section 2 outlines the FH-based barycentric rational interpolation; Section 3 demonstrates practical construction of approximation models through some numerical examples; Section 4 discusses the quality of the approximants obtained in Section3; and Section 5 concludes.

2 Rational interpolation Method

2.1 Barycentric rational interpolation

Given a function $f : \mathbb{R}^N \to \mathbb{R}$ and the interpolation data $\{(x_i, f(x_i))\}_{i\in I}$, rational interpolation for f is a problem to find a rational function $r : \mathbb{R}^N \to \mathbb{R}$ satisfying $f(x_i) = r(x_i), \forall i \in I$ which gives a good approximation to f. Throughout this study, we assume that the interpolation nodes are structured as a tensor-product grid (Cartesian grid) in $\mathbb{R}^N$.

As a widely accepted standard approach, multi-variate barycentric rational interpolation[2] takes the following form, known as the "barycentric form":

$$r(x) := \sum_{i_1=1}^{n_1} \cdots \sum_{i_N=1}^{n_N} f(x_{i_1}^{(1)}, \ldots, x_{i_N}^{(N)}) \prod_{j=1}^{N} \frac{\frac{w_{i_j}^{(j)}}{x^{(j)} - x_{i_j}^{(j)}}}{\sum_{k=1}^{n_j} \frac{w_k^{(j)}}{x^{(j)} - x_k^{(j)}}} \tag{1}$$

where $x = (x^{(1)}, \ldots, x^{(N)})^T$, and $w_{i_j}^{(j)}$ $(1 \le i_j \le n_j, 1 \le j \le N)$ are parameters that determine the approximant $r(x)$. For convenience, we often use the vectorized notation $w^{(j)} := (w_1^{(j)}, \ldots, w_{n_j}^{(j)})^T$ for the j-th variate.

The question is how to find a better set of interpolation nodes $\{x_i\}_{i\in I}$ and $\{w^{(j)}\}_{1\le j\le N}$ which give good approximation model r of f. In this paper, we employ a prominent method for barycentric rational interpolation: the Floater-Hormann method.

2.2 Floater-Hormann method

For the sake of simplicity, let $N = 2$. Floater-Hormann method [1] (FH method) gives $\{w^{(s)}\}_{s=1,2}$ in the following form with given interpolation nodes $\{(x_i^{(1)}, x_j^{(2)})\}_{1\le i\le n_1, 1\le j\le n_2}$

$$w_k^{(s)} := (-1)^{k-d_s} \sum_{i=\max\{0,k-d_s\}}^{\min\{k,n_s-d_s\}} \prod_{j=i, j\neq k}^{i+d_s} \frac{1}{|x_k^{(s)} - x_j^{(s)}|}$$

where $d = (d_1, d_2)$ is called "blending parameter", a pair of integers satisfying $0 \le d_s \le n_s$. If f is a sufficiently smooth function, the convergence rate is as follows ([3]):

$$\|f - r\|_\infty = O(h_1^{d_1+1} h_2^{d_2+1}),$$
$$h_1 := \max_{1\le i\le n_1-1} |x_{i+1}^{(1)} - x_i^{(1)}|, \quad h_2 := \max_{1\le i\le n_2-1} |x_{i+1}^{(2)} - x_i^{(2)}|.$$

A key advantage of the FH method is that it ensures the approximant r is free from spurious poles [1, 3, 4], which frequently occur in traditional rational interpolation methods.

3 Application to the inverse transform sampling

In this section, we show practical application of FH method to constructing approximation model of inverse CDFs necessary for the inverse transform sampling through several numerical examples. First of all, we would like to clarify our strategy and aim of the model construction process.

Data Generation: To construct the interpolation data, we utilize Brent's root-finding algorithm. For a given distribution, the grid points are numerically computed by solving the equation $F_X(x) - u_i = 0$ to obtain $x_i = F_X^{-1}(u_i)$. Similarly, the conditional inverse CDF data are collected as $\{(v_i, x_j, F^{-1}_{Y|X=x_j}(v_i))\}$. These calculated values are treated as the true reference for the interpolation process.

Notaions: We often denote the target inverse CDFs as $G_1(u) := F_X^{-1}(u), G_2(v, x) := F^{-1}_{Y|X=x}(v)$ and their corresponding approximants as r_1, r_2.

Sampling Domain: We suppose it is sufficient to approximate the target inverse CDFs, $G_1(u)$ and $G_2(v, x)$, within the strictly bounded domain $u, v \in [\epsilon, 1 - \epsilon]$ where $\epsilon = 10^{-8}$. This parameter choice is directly motivated by the coverage characteristics of practical QMC sequences. In large-scale financial simulations employing up to $M = 10^7$ samples, low-discrepancy sequences such as Sobol' sequences typically generate extreme values no closer than $\frac{1}{2M} \approx 5 \times 10^{-8}$ to the boundaries. Therefore, setting $\epsilon = 10^{-8}$ ensures that all generated uniform samples fall entirely within the evaluation domain, safely avoiding any need for out-of-domain extrapolation.

Target Accuracy: The statistical error of QMC utilizing typical sequences like Sobol' sequence has the convergence rate of $O(\frac{(\log M)^s}{M})$, where s is dimensionality[5]. In typical financial applications, the empirical statistical error of QMC methods with $M = 10^7$ samples generally plateaus around the 10^{-6} to 10^{-5} level. To ensure that our surrogate model does not become the dominant source of error, we target an approximation accuracy of 10^{-7} order. This strict tolerance keeps the approximation error well below the expected QMC statistical error, guaranteeing negligible impact on subsequent option pricing and risk metrics.

Transformation of variables: The target inverse CDFs often exhibit ill behavior like steep gradients near the boundaries, which can degrade interpolation accuracy on simple equidistant grids. To address this, we apply appropriate variable transformations, $u \to s = s(u)$ and $x \to t = t(x)$, and construct the rational interpolants in the substituted coordinate space. This pre-processing effectively smoothes out sharp variations, drastically decreasing the number of nodes necessary for accurate approximation and ensuring computational efficiency in the online phase.

3.1 Example #1

$$f_X(x) = \begin{cases} \frac{1}{\pi\sqrt{x(1-x)}}(1+\sin(8\pi x)) & (0 < x < 1) \\ 0 & (\text{otherwise}) \end{cases}$$

Target: $G_1(u)$ on $u \in [10^{-8}, 1-10^{-8}]$

Transformation:

$$s(u) = \arcsin(2u-1) + 0.5\sum_{k=1}^{4}\sqrt[3]{u-u_k}$$

where $u_k = F_X(\frac{4k-1}{16})$ for $k = 1, 2, 3, 4$

Nodes: $n_1 = 320$, equidistant

Parameter: $d_1 = 3$.

Accuracy: $\|G_1 - r_1\|_\infty < 4.93 \times 10^{-8}$

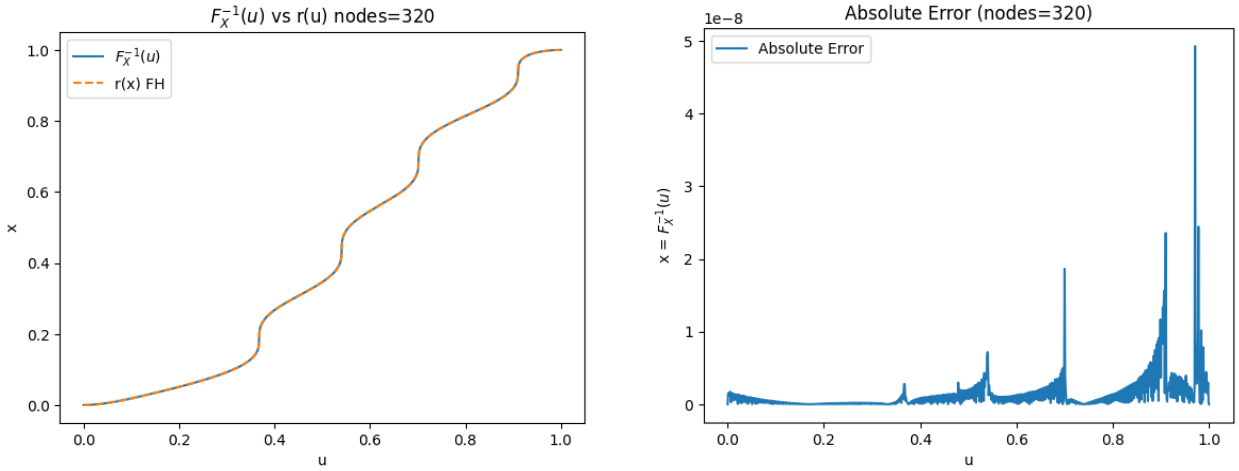


Figure 2: Ex.1: $G_1(u)$ and Absolute Error

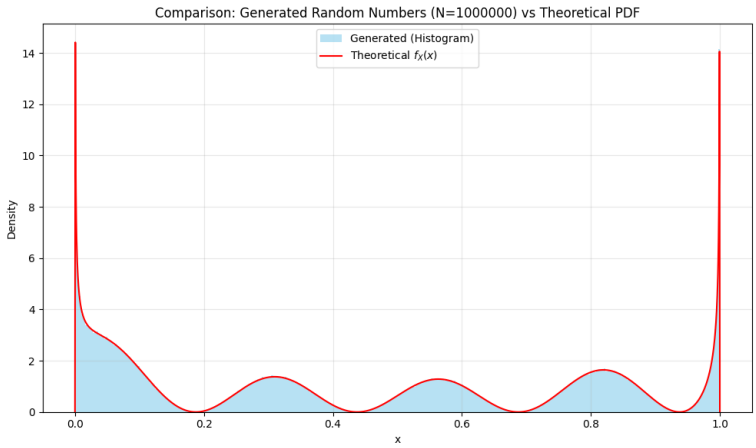


Figure 3: Ex.1: Histogram and $f_X(x)$

3.2 Example #2

$$f_{X,Y}(x,y) = c\left(e^{-x^2-y^2} + \frac{1}{2}e^{-(x-2)^2-(y-2)^2}\right), \quad (x,y) \in \mathbb{R}^2 \tag{2}$$

Target1: $G_1(u)$ on $u \in [10^{-8}, 1-10^{-8}]$

Transformation: $s = \Phi^{-1}(u)$ ($\Phi(x)$ is the CDF of the normal distribution)

Nodes: $n_1 = 145$, equidistant

Parameter: $d_1 = 3$

Accuracy: $\|G_1 - r_1\|_\infty < 8.69 \times 10^{-8}$

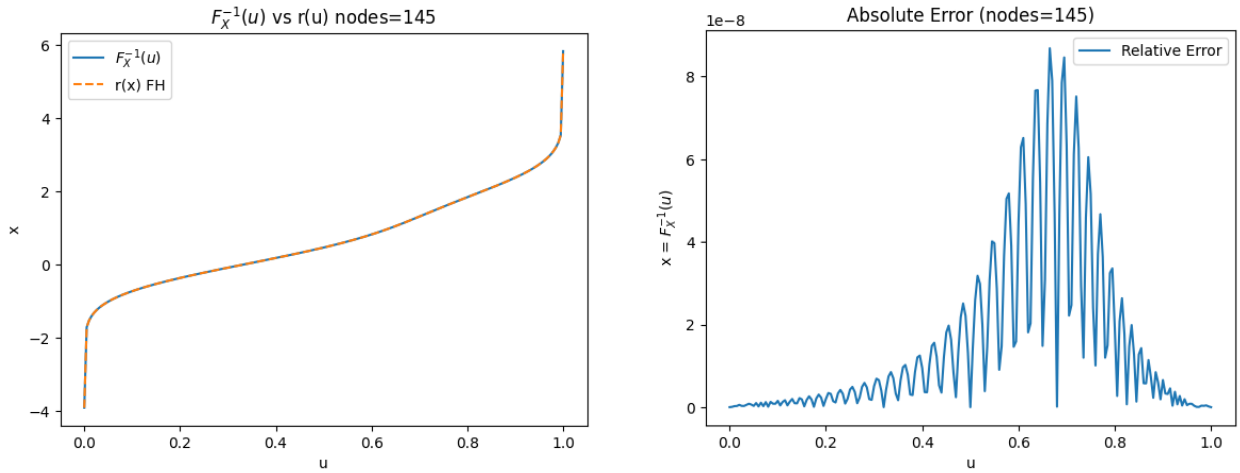


Figure 4: Ex.2: $G_1(u)$ and Absolute Error

Target2: $G_2(v,x)$ on $(v,x) \in [10^{-8}, 1-10^{-8}] \times \mathbb{R}$

Transformation: $s = \Phi^{-1}(v)$, $t = \frac{2}{1+e^{-0.5(x-1.173)}}$

Nodes: $(n_1, n_2) = (150, 210)$, equidistant

Parameter: $(d_1, d_2) = (5, 5)$

Accuracy: $\|G_2 - r_2\|_\infty < 7.91 \times 10^{-8}$

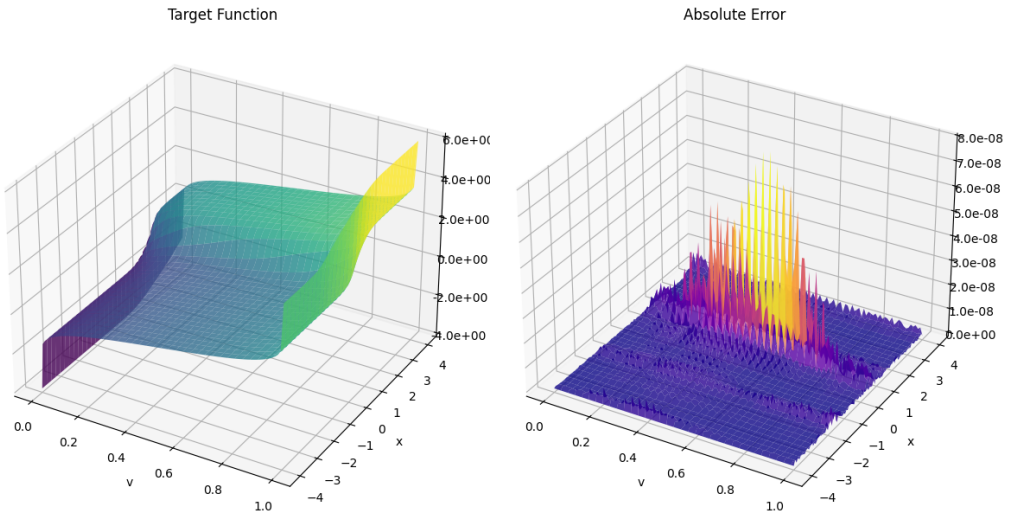


Figure 5: Ex.2: $G_2(v, x)$ and Absolute Error

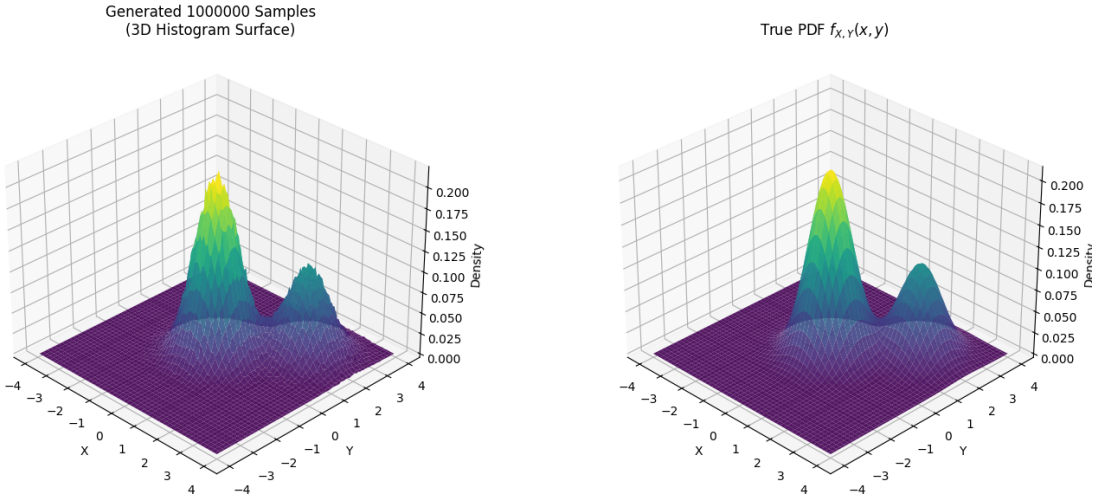


Figure 6: Ex.2: Histogram and $f_{X,Y}(x, y)$

3.3 Example #3

$$f_{X,Y}(x,y) = c\frac{\exp(-0.2\cdot x)}{\left(1+x^2+(y-\log(1+x))^2\right)^2}, \qquad (x,y)\in[0,20]\times[-5,10] \tag{3}$$

Target1: $G_1(u)$ on $u \in [10^{-8}, 1-10^{-8}]$

Transformation: $s = \log(\frac{u}{1-u})$

Nodes: $n_1 = 70$, equidistant

Parameter: $d_1 = 5$

Accuracy: $\|G_1 - r_1\|_\infty < 2.23 \times 10^{-7}$

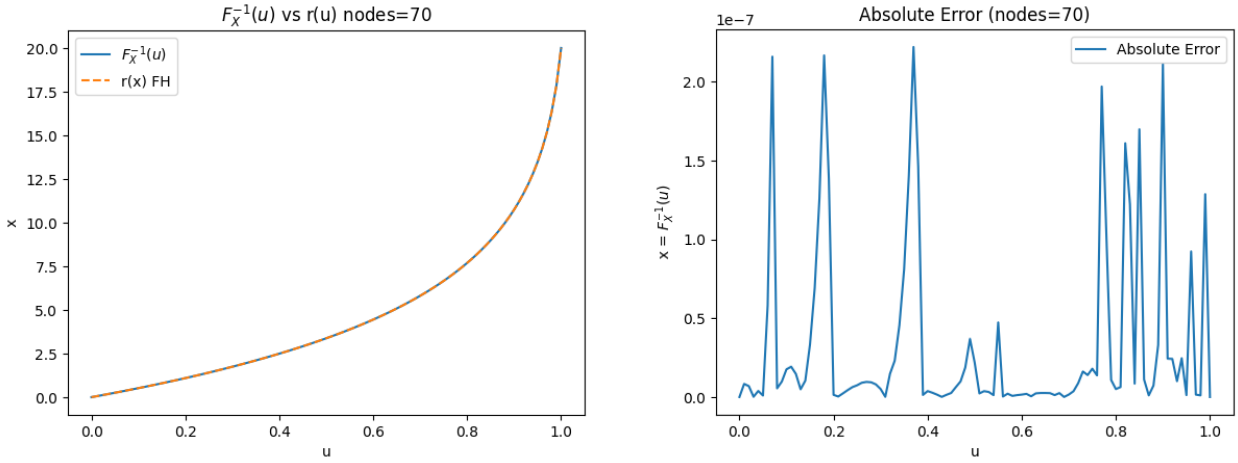


Figure 7: Ex.3: $G_1(u)$ and Absolute Error

Target2: $G_2(v,x)$ on $(v,x) \in [10^{-8}, 1-10^{-8}] \times [0,20]$

Transformation: $s = \log(\frac{v}{1-v})$, $t = \log(0.5+x)$

Nodes: $(n_1, n_2) = (200, 200)$, equidistant

Parameter: $(d_1, d_2) = (3, 3)$.

Accuracy: $\|G_2 - r_2\|_\infty < 2.02 \times 10^{-7}$

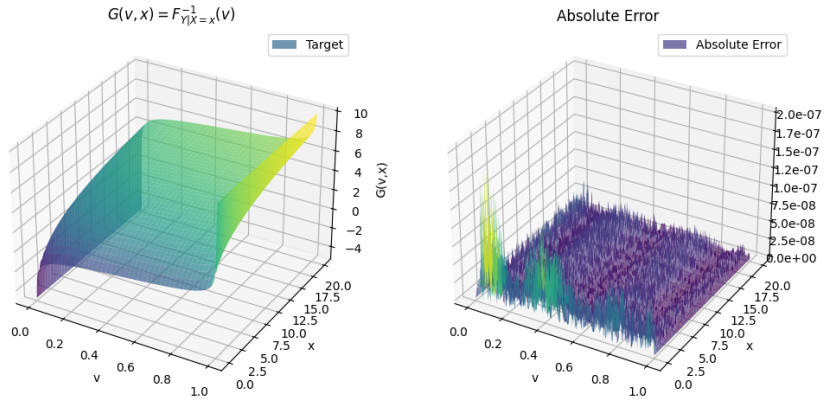


Figure 8: Ex.3: $G_2(v, x)$ and Absolute Error

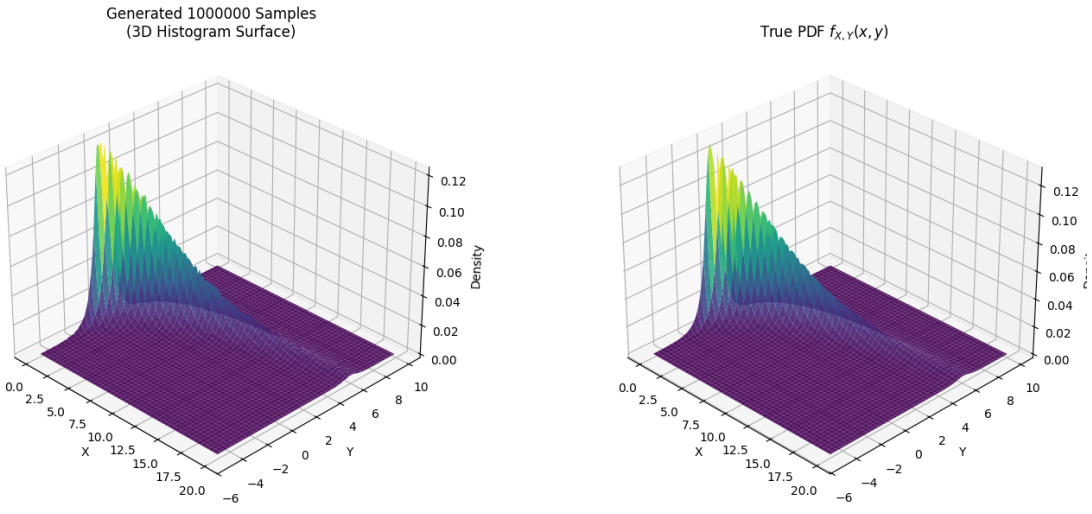


Figure 9: Ex.3: Histogram and $f_{X,Y}(x, y)$

4 Discussion

4.1 Comparison with Sequential Root-Finding

To clarify the practical significance of the proposed rational surrogate approximation model, we evaluate its computational efficiency against the conventional approach: the sequential execution of Brent's root-finding algorithm for every sample generation.

Table 1 summarizes the computational time required to generate $10,000$ joint random samples.

Table 1: Computational time (sec) for generating 10,000 samples.

Example	Brent's root-finding	Rational model
#1	868.426	0.063
#2	1413.098	0.108
#3	5100.045	0.085

Since the baseline Brent's method inherently requires numerical integration to evaluate the non-analytical CDFs for every single sample generation—including handling sharp boundary singularities (as in Ex. #1) or executing a nested integration loop for conditional joint distributions (as in Ex. #2 and #3)—it inevitably scales the execution time to a prohibitive level.

In contrast, the proposed rational surrogate model bypasses both numerical integration and root-finding during the online sampling phase across all examples, achieving consistently ultra-fast performance.

It is worth emphasizing that under the strict constraint where "the inverse transform sampling must be utilized to preserve QMC properties," general-purpose root-finding algorithms like Brent's method are widely used. By shifting the heavy computational burden of root-finding and numerical integration entirely to the offline data-generation phase, the proposed barycentric rational interpolation framework provides an overwhelmingly fast, real-time sampling performance in the online phase. This capability grants the proposed method a decisive practical value in financial engineering.

4.2 Convergence of QMC Error

To evaluate the numerical accuracy and convergence behavior, we performed QMC and standard MC simulations to estimate the expectation $\mathbb{E}[XY]$ under Example #3. Figure 10 illustrates the integration error as a function of the sample size M, ranging from $2^{10} = 1{,}024$ to $2^{24} = 16{,}777{,}216$. The estimates derived from the proposed surrogate model (red) align perfectly with those computed using Brent's root-finding method (blue) across all sample sizes. This exact agreement confirms that our targeted target accuracy of 10^{-7} introduces no noticeable degradation to QMC integration. Furthermore, Figure 10 demonstrates that the QMC scheme maintains a faster convergence rate ($\approx O(M^{-1})$) compared to the standard MC approach ($O(M^{-1/2})$).

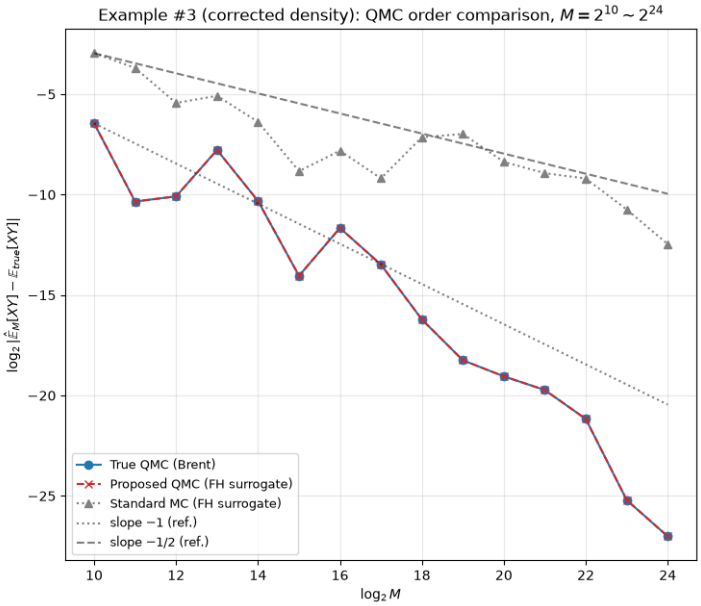


Figure 10: sample size M VS error of QMC

5 Conclusion

In this study, we demonstrated construction process of surrogate models of inverse CDFs for the inverse transform sampling, utilizing the barycentric rational interpolation based on the Floater-Hormann method. Through several experiments, we demonstrated that the proposed method achieves high approximation accuracy and it worked in the context of a practical use of QMC method. While the present approach relies on ad hoc variable transformations derived through manual trial-and-error procedures to achieve optimal approximation accuracy, automating this process is a highly promising direction. One approach is to define a "monitor function" $M(u)$ which detects the positions of steep gradiants in the domain of the inverse CDF and automatically provides appropriate transformation map $s(u)$. If this makes good, it would be a big step for a versatile approximation framework for inverse CDF, making the inverse transform sampling a much more versatile tool for a random number generation.

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